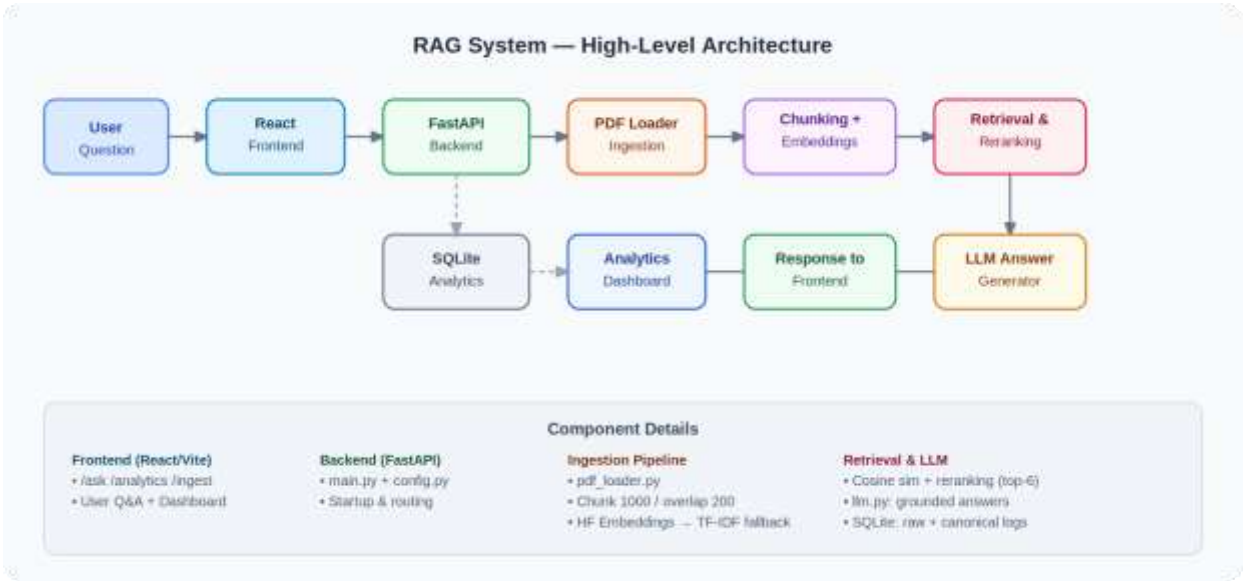


Architecture Overview

1. Flow

The diagram below illustrates the end-to-end data flow from user question to grounded answer and analytics logging.



The pipeline operates as follows:

1. The frontend sends questions to the FastAPI backend.
2. The backend checks whether the PDF has been ingested.
3. If not, the ingestion flow extracts text from the PDF, chunks it, and stores embeddings.
4. The retrieval layer ranks the most relevant chunks for the question.
5. The answer-generation layer uses the retrieved chunks to produce a grounded answer.
6. Analytics logs are written to SQLite so the dashboard can show query trends.

2. Core Components

The table below maps each component to its source file and primary responsibility.

Component	File / Module	Description
Frontend	React/Vite	User Q&A interface; calls /ask, /analytics, and /ingest endpoints.

Component	File / Module	Description
Backend	backend/app/main.py	FastAPI app handling API routes and startup initialization.
Config	backend/app/config.py	Configuration loader consumed by the backend.
PDF Loader	backend/app/pdf_loader.py	Extracts text from PDF and triggers chunking.
Chunker	LangChain splitter	RecursiveCharacterTextSplitter; chunk 1000, overlap 200.
Embeddings	HF / TF-IDF	Hugging Face embeddings preferred; TF-IDF fallback for simple envs.
Retrieval	Cosine similarity	Top-k retrieval with query boosts and reranking.
LLM	backend/app/llm.py	Builds prompt context, calls LLM provider, applies grounded fallbacks.
Analytics DB	backend/app/database.py	Stores raw query logs and canonical grouped queries.

3. Data Flow Example

User asks: *"What services are covered by the agreement?"*

- The question is embedded into a vector representation.
- Relevant chunks are retrieved from the indexed PDF using cosine similarity.
- The answer logic synthesizes a grounded response from the retrieved evidence.
- The answer is returned to the React frontend.
- The interaction is logged to SQLite for analytics.

4. Design Decisions

Aspect	Decision & Rationale
Chunk size	1000 characters with 200 overlap preserves clause context across boundaries.
Top-k retrieval	Default 6 results with rerank pool expansion for better candidate selection.
Embeddings	Hugging Face embeddings preferred; TF-IDF fallback for simple/offline environments.
Answer style	Prioritize direct, concise answers over long verbatim excerpts.

Aspect	Decision & Rationale
Analytics	Store both raw query text and canonical grouped queries for richer reporting.